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Roll No

IT-403 (GS)

B.E. IV Semester Examination, June 2020

Grading System (GS)

Data Base Management System

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. What is BCNF? Give an example of a relation schema R and set of dependencies such that R is in BCNF, but not in UNF.
2. What is Canonical cover? Consider Following set F of functional dependencies on schema R(A, B, C) and compute Canonical cover for F.
{A→BC, B→C, A→B, AB→C}
3. Write the full form of the following :
 - i) SQL
 - ii) QUEL
 - iii) QBE
 - iv) DBA
 - v) E-R diagram
 - vi) DBMS
 - vii) RDBMS
4. Explain the following term
 - a) Instances
 - b) Primary key
 - c) Tuples
 - d) Domains
 - e) Schemas
5. What is union compatibility? What are the relational algebra operators that require the relations on which they are applied be union compatible?

OR

 - a) Discuss the various components of database system.
 - b) What are the different levels of data abstraction? Explain each of them.
6.
 - a) What is Normalization? Explain 1NF and 2NF with taking a suitable example.
 - b) What is Functional dependency?
7. Write short note (any two).
 - a) Data Independence.
 - b) Dependency preservation.
 - c) Intension and extension.
8. Write a short notes (any three) :
 - i) Multi valued dependencies
 - ii) Loss less join
 - iii) Data Models
 - iv) Views
 - v) Transaction processing
